

1. Core Concepts & Modelling

LTi state-space:

$$\dot{\mathbf{x}} = \mathbf{A}\mathbf{x} + \mathbf{B}u + \mathbf{D}, \quad y(t) = \mathbf{C}\mathbf{x}$$

A=dynamics (main), B=input, C=output, D=disturbance. $A \in \mathbb{R}^{n \times n}$, $B \in \mathbb{R}^{n \times 1}$, $C \in \mathbb{R}^{1 \times n}$. Feedthrough form: $y = \mathbf{C}\mathbf{x} + \mathbf{D}u$.

ODE $\rightarrow$ **SS**: for $\ddot{x} + a\dot{x} + bx = u$, $\mathbf{x} = (x, \dot{x})^T$:

$$A = \begin{pmatrix} 0 & 1 \\ -b & -a \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad C = (1 \quad 0)$$

3rd order $\ddot{x} + a\ddot{x} + b\dot{x} + cx = u$, $\mathbf{x} = (x, \dot{x}, \ddot{x})^T$:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -c & -b & -a \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad C = (1 \quad 0 \quad 0)$$

TF $\rightarrow$ **SS**: for $G(s) = \frac{b_0}{s^2 + a_1s + a_0}$:

$$A = \begin{pmatrix} 0 & 1 \\ -a_0 & -a_1 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad C = (b_0 \quad 0)$$

3rd order $G(s) = \frac{b_0}{s^3 + a_2s^2 + a_1s + a_0}$:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -a_0 & -a_1 & -a_2 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \quad C = (b_0 \quad 0 \quad 0)$$

Mass-spring-damper: $m\ddot{y} + c\dot{y} + ky = u$, $G(s) = \frac{1}{ms^2 + cs + k}$.

2. Linearisation (Lec 2)

For $\dot{x} = f(x, u)$:

(1) Equilibrium $f(x_e, u_e) = 0$.

(2) Taylor (1st order):

$$\dot{x} \approx f(x_e, u_e) + \left. \frac{\partial f}{\partial x} \right|_e (x - x_e) + \left. \frac{\partial f}{\partial u} \right|_e (u - u_e)$$

General (scalar) form: $F(v_0 + \delta v) \approx F(v_0) + \left. \frac{\partial F}{\partial v} \right|_{v_0} (v - v_0)$.

(3) Deviation vars $\delta x = x - x_e$, $\delta u = u - u_e$:

$$\dot{\delta x} \approx \left. \frac{\partial f}{\partial x} \right|_e \delta x + \left. \frac{\partial f}{\partial u} \right|_e \delta u$$

Multivariable: $A = \left. \frac{\partial f}{\partial \mathbf{x}} \right|_e$ (Jacobian), $B = \left. \frac{\partial f}{\partial u} \right|_e$.

3. First-Order Systems (Lec 3)

$\dot{x} + ax = u$, free resp.: $x(t) = e^{-at}x(0)$.

Stability: pole at $s = -a$. $a > 0 \Rightarrow$ decays, **stable**; $a < 0 \Rightarrow$ grows, **unstable**.

Step ($u = \text{const}$), $x(0) = x_0$:

$$x(t) = \left(x_0 - \frac{u}{a}\right)e^{-at} + \frac{u}{a}$$

Steady state: $x(\infty) = u/a$. Time constant $\tau = 1/a$.

Time specs: rise $t_r \approx \frac{2.2}{a} = 2.2\tau$ (10–90%); settle $t_s \approx \frac{3.91}{a} \approx \frac{4}{a} = 4\tau$ (2%).

Transfer function: $G(s) = \frac{\text{output}}{\text{input}} = \frac{Y(s)}{U(s)}$, $y = G(s)u$, $u = Ue^{st}$.

$G(s) = \frac{X(s)}{U(s)}$. For $\ddot{x} + 3\dot{x} + 2x = u$: $G = \frac{1}{s^2 + 3s + 2}$.

DC gain $\rightarrow$ **final value**: for any stable G , step of size U :

$$x(\infty) = G(0) \cdot U \quad (\text{set all derivs} = 0).$$

$G(0)$ sign = direction ($G(0) < 0 \Rightarrow$ output dips below 0); magnitude = final value. Generalises $x_\infty = u/a$.

4. Proportional / Steady-State Error

Plant $\dot{x} + ax = u$, $u = K_P(r - x)$:

$$\dot{x} + (a + K_P)x = K_P r$$

$x(\infty) = \frac{K_P r}{K_P + a}$, error $e(\infty) = r \frac{a}{K_P + a}$.

With disturbance d , noise n :

$$x_{ss} = \frac{K_P(r - n) + d}{a + K_P}$$

Large $K_P \Rightarrow$ small error (**feedback stabilisation**: pick $K_P > -a$). **Integral** action $\Rightarrow$ zero steady-state error.

SS error (unity fb, unit step): $e(\infty) = \frac{1}{1 + G(0)}$ ($G(0)$ =DC loop

gain; position const $K_P = G(0)$).

System type: # poles of $L = CG$ at $s=0$: 0, 1, 2 $\Rightarrow$ Type 0, 1, 2.

	Type 0	Type 1	Type 2
Step	$\frac{1}{1 + K_P}$	0	0
Ramp	∞	$\frac{1}{K_v}$	0
Parab.	∞	∞	$\frac{1}{K_a}$

5. Homogeneous Solns & Poles (Lec 4)

Char. eq. of $\ddot{x} + a\dot{x} + bx = 0$: $\lambda^2 + a\lambda + b = 0$.

Real distinct: ($a^2 - 4b > 0$): $x_h = C_1 e^{\lambda_1 t} + C_2 e^{\lambda_2 t}$.

Real repeated: ($a^2 - 4b = 0$): $x_h = (C_1 + C_2 t) e^{\lambda^* t}$.

Complex: $\lambda = -\sigma \pm j\omega_d$:

$$x_h = e^{-\sigma t} (C_3 \cos \omega_d t + C_4 \sin \omega_d t)$$

Stability: all poles (roots of denom.) have negative real part $\Leftrightarrow x(t) \rightarrow 0$ for any $x(0)$, zero input.

Necessary cond.: char. poly has *all coefficients present & positive* (same sign) — any missing/sign change $\Rightarrow$ unstable. Real, distinct, negative poles $\Rightarrow$ stable & non-oscillatory.

$G(s) = \frac{b_1 s + b_0}{s^2 + a_1 s + a_0}$; **poles**=denom roots, **zeros**=numer roots.

6. Second-Order Systems (Lec 5)

Standard: $G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$.

Poles $s = -\zeta\omega_n \pm j\omega_n \sqrt{1 - \zeta^2} = -\sigma \pm j\omega_d$.

$$\sigma = \zeta\omega_n, \quad \omega_d = \omega_n \sqrt{1 - \zeta^2}$$

$$\omega_n = \sqrt{\sigma^2 + \omega_d^2}, \quad \zeta = \sigma/\omega_n$$

ζ = sine of angle from imag. axis. Underdamped: $0 < \zeta < 1$.

Envelope: $x(t) = e^{-\sigma t} \sin(\omega_d t)$.

Response specs: $M_p = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$ (overshoot), $t_p = \frac{\pi}{\omega_d}$ (peak time),

$t_s \approx \frac{4}{\zeta\omega_n}$ (2% settle), $t_r \approx \frac{1.8}{\omega_n}$ (rise 10–90%).

7. PID Control (Lec 5)

$$C(s) = K_P + \frac{K_i}{s} + K_d s = \frac{K_d s^2 + K_P s + K_i}{s}$$

• **P**: fast, may leave SS error / oscillate.

- **PI**: removes SS error, slower transient.
- **PD**: adds damping, faster, noise-sensitive, adds a zero.
- **PID**: removes SS error + improves transient.

$K_P \uparrow \Rightarrow \omega_n \uparrow$ (faster), $\zeta \downarrow$ (more overshoot).

Rate feedback: $u = K_P(r - x) - K_D \dot{x}$ (avoids derivative kick):

$$G(s) = \frac{K_P + K_D s}{s^2 + (1 + K_D)s + K_P}$$

zero at $s = -K_P/K_D$.

PD design ex. $\ddot{x} + 3\dot{x} + 2x = u$, $u = K_P(r - x) - K_d \dot{x}$:

$$G = \frac{K_P}{s^2 + (3 + K_d)s + (K_P + 2)}$$

match to $s^2 + 2\zeta\omega_n s + \omega_n^2$.

9. State-Space Response (Lec 6)

Matrix exponential: $e^{At} = \sum_{k=0}^{\infty} \frac{(At)^k}{k!} = I + At + \frac{(At)^2}{2!} + \dots$

Free resp.: $\mathbf{x}(t) = e^{At}\mathbf{x}(0)$.

Forced: $\mathbf{x}(t) = e^{At}\mathbf{x}_0 + \int_0^t e^{A(t-\tau)} \mathbf{B}u \, d\tau$.

TF from SS: $G(s) = C(sI - A)^{-1}B + D$.

Output to $u = Ue^{st}$: $y = Ce^{At}\mathbf{x}_0 + G(s)Ue^{st}$.

Stable $\Leftrightarrow$ all eigenvalues of A have $\text{Re} < 0$.

$e^{At} = Te^{Jt}T^{-1}$ (Jordan); repeated eig give t^k terms.

10. Reachability / Control (Lec 7)

Reachability matrix: (3rd order: 3×3)

$$W = \begin{bmatrix} B & AB & A^2B \end{bmatrix}$$

Reachable $\Leftrightarrow \det W \neq 0$.

Equivalent: can place *any* closed-loop poles via K .

State feedback: $u = -K\mathbf{x}$, closed loop $\dot{\mathbf{x}} = (A - BK)\mathbf{x}$.

Design: $\det(sI - (A - BK)) \stackrel{!}{=} \text{desired char. poly, match coefficients}$.

Transformation: $T = W\tilde{W}^{-1}$ ($\tilde{W}$ from reachable canonical form), $K = \tilde{K}T$.

Reachable canonical form: for $\ddot{y} + a_1\dot{y} + a_2y = u$ (top-row companion):

$$\tilde{A} = \begin{pmatrix} - & -a_2 \\ 1 & 0 \end{pmatrix}, \quad \tilde{B} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

char. poly $s^2 + a_1s + a_2$ read off top row.

Dominant poles: choose 2nd-order pair for transient; place rest 5–10 $\times$ further left, low overshoot.

11. Observability / Observers (Lec 8)

Observability matrix: (3rd order: 3×3)

$$W_o = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}$$

Observable $\Leftrightarrow \det W_o \neq 0$: recover $\mathbf{x}(0)$ from u, y on $[0, T]$.

Luenberger observer:

$$\dot{\hat{\mathbf{x}}} = \mathbf{A}\hat{\mathbf{x}} + \mathbf{B}u + \mathbf{L}(y - \mathbf{C}\hat{\mathbf{x}})$$

Error $\tilde{\mathbf{x}} = \mathbf{x} - \hat{\mathbf{x}}$: $\dot{\tilde{\mathbf{x}}} = (\mathbf{A} - \mathbf{L}\mathbf{C})\tilde{\mathbf{x}}$.

Condition: $A - LC$ stable ($\Rightarrow \tilde{\mathbf{x}} \rightarrow 0$). Roots (eig) of $A - LC$: **negative** $\Rightarrow$ error decays, $\hat{\mathbf{x}} \rightarrow \mathbf{x}$ (convergent); **positive** $\Rightarrow$ unstable, error grows, never converges.

Design: $\det(sI - (A - LC)) \stackrel{!}{=} \text{desired poly, match coeffs}$.

Output feedback: (separation) observer + $u = -K\hat{\mathbf{x}} + Nr$. CL char poly: $\det(sI - (A - BK)) \det(sI - (A - LC))$.

Separation principle: applies to systems that are *both reachable &*

observable: design K and L **independently**; the CL poles are just the union of the controller poles (eig $A-BK$) and observer poles (eig $A-LC$).

Feedforward: (zero SS err): $N = -(C(A-BK)^{-1}B)^{-1}$.
Integral action: augment state $\dot{q} = Cx - \bar{y}$ to kill SS error without exact model.

12. Frequency Response (Lec 9)

Input $A \sin \omega t \Rightarrow y_{ss} = A|G(j\omega)| \sin(\omega t + \phi)$
 $G(j\omega) = |G(j\omega)|e^{j\phi}$, $\phi = \angle G = \arctan \frac{\text{Im}}{\text{Re}}$.
 $\angle G = \sum \angle(\text{num}) - \sum \angle(\text{den})$: **numerator** (zeros) add +ve phase, **denominator** (poles) add -ve phase.
Complex algebra: $z_1 z_2$: multiply mags, add phases; z_1/z_2 : divide mags, subtract phases.
Ex. $G = \frac{1}{j\omega(j\omega+1)(j\omega+3)}$:

$$|G| = \frac{1}{\sqrt{\omega^2(\omega^2+1)(\omega^2+9)}}$$
$$\angle G = -\left(\frac{\pi}{2} + \arctan \omega + \arctan \frac{\omega}{3}\right)$$

Forced sinusoid (undetermined coeffs): for $\ddot{x} + \omega_n^2 x = \sin \omega t$, try $x = X \sin \omega t$, match:

$$X = \frac{1}{\omega_n^2 - \omega^2}$$

real $\Rightarrow$ phase $0^\circ/180^\circ$ (no lag); $X \rightarrow \infty$ as $\omega \rightarrow \omega_n$ (**resonance**, undamped).

13. Bode Plots (Lec 9-10)

Mag in dB: $20 \log_{10} |G(j\omega)|$. $K = 10^{dB/20}$, $dB = 20 \log_{10} K$. $\times 10 \Rightarrow 20\text{dB}$; $1 \rightarrow 0\text{dB}$. **Standard form:** factor so each bracket starts with 1; constant out front = DC gain K :

$$G(s) = \frac{K(\frac{s}{a}+1)}{s^n(\frac{s}{b}+1)}$$

Start height (flat, low ω) = $20 \log_{10} K$ dB. If sloped (i.e pole @ 0), choose small ω and S.H. = $20 \log_{10} \left(\frac{K}{\omega}\right) \Rightarrow K = \omega \cdot 10^{\text{S.H./20}}$.

Mag slopes: (added at break freq $\omega= p , z $):	
DC gain K	flat at $20 \log K$
integrator $1/s^n$	$-20n$ dB/dec from $\omega \rightarrow 0$
diff. s	$+20$ dB/dec
pole $(\frac{s}{b}+1)^{-1}$	-20 dB/dec after $\omega=b$
zero $(\frac{s}{a}+1)$	$+20$ dB/dec after $\omega=a$
cplx poles @ ω_n	-40 dB/dec; peak $M_z = \frac{1}{2\zeta\sqrt{1-\zeta^2}}$ at ω_n
cplx zeros @ ω_n	$+40$ dB/dec

tion ± 1 decade about corner.	Phase: $\angle G = \sum \angle(\text{num}) - \sum \angle(\text{den})$; each $(j\omega+a)$: $\arctan \frac{\omega}{a}$; transi-	DC gain	0°
		integrator $1/s$	-90° (all ω)
		diff. s	$+90^\circ$ (all ω)
		pole @ b	-90° total, centred b
		zero @ a	$+90^\circ$ total, centred a
		cplx poles @ ω_n	-180° total

Bode $\rightarrow$ TF: (1) DC gain $K=10^{X\text{dB}/20}$ from flat low- ω ($\omega=1$ if slanted

start). (2) initial slope: flat=no changes, $-20/\text{dec}=\text{one } 1/s$, $-40=1/s^2$, $+20=s$. (3) corners where slope *changes*: extra $-20 \Rightarrow$ pole, $+20 \Rightarrow$ zero, $\mp 40 \Rightarrow$ cplx pair @ ω_n . (4) phase confirms: start $0/-90/-180^\circ=0/1/2$ integrators; drop @ corner=pole, rise=zero. (5) assemble:

$$G(s) = \frac{K s^{\text{pos}}(s+z_1)(s+z_2) \cdots}{s^{\text{int}}(s+p_1)(s+p_2) \cdots}$$

Sanity: final slope = $-20(\#p-\#z)$ dB/dec; final phase = $-90^\circ(\#p-\#z)$; DC check: sub $s=0$.
GM & PM on Bode: GM = magnitude (dB) at phase = -180° ; PM = phase (deg) at magnitude 0 dB.

14. Nyquist Criterion (Lec 10,12)

Plot $L(j\omega)$, $\omega: -\infty \rightarrow \infty$. Closed loop $G_{yr} = \frac{L}{1+L}$.
Theorem: P =open-loop RHP poles of L ; N =net *clockwise* encirclements of -1 . Then CL has $Z = N + P$ RHP poles.
Stable $\Leftrightarrow Z = 0 \Leftrightarrow N = -P$ (P anticlockwise encirclements).
Stable- L case: stable iff *no* encirclements of -1 .
CL poles satisfy $L(s) = -1$.
Real-axis crossing: Nyquist evaluates G along the imag. axis, so to find where it crosses $(\alpha, 0)$: set $G(s)=\alpha$ and solve — the root is **purely imaginary** $s = \pm j\omega$.

15. Stability Margins (Lec 10,11)

Crossover ω_{gc} : $|L(j\omega_{gc})| = 1$ (0 dB).
Gain margin: $g_m = 1/|L(j\omega_{pc})|$ at phase = -180° . Find where Nyquist crosses -ve real axis; GM = inverse of that magnitude. (+ve).
Phase margin: $\varphi_m = 180^\circ + \angle L(j\omega_{gc})$. Find where Nyquist hits unit circle; PM = angle from that point to -ve x-axis.
Stability margin: s_m =closest dist of Nyquist to -1 ; $|S(j\omega)| = 1/\text{dist to } (-1)$. Shortest distance from Nyquist curve to $(-1, 0)$.
 $|S(j\omega)|$ = sensitivity.
 $PM = 2 \sin^{-1} \left(\frac{1}{2|S(j\omega_c)|} \right)$.
Time delay: Max delay $\tau_{max} = \varphi_m/\omega_{gc}$.

16. Sensitivity / Gang of Four (Lec 11)

$$S = \frac{1}{1+PC}, \quad T = \frac{PC}{1+PC}, \quad S+T=1$$

$S+T=1$: a seesaw — low S (tracking/dist. rejection) vs. low T (robust to noise/model uncertainty); shape each small in different bands.
3 TFs: $d, r, n.$ $\rightarrow$ Set other 2 to 0 and solve.
Disturbance: $\eta = PSd$. $|S|<1$ attenuates; $|S|=0$ eliminates; $|S|>1$ worsens.
 S measures sensitivity of CL to plant uncertainty.
Loop shaping — why: shape $L=PC$ to balance performance vs stability.
how: (i) high gain / integrator at low $\omega \rightarrow$ tracking + reject d ($S \approx 0$); (ii) low gain + steep roll-off at high $\omega \rightarrow$ reject noise n ($T \approx 0$); (iii) gentle slope ($\approx -20\text{dB/dec}$) near $\omega_{gc} \rightarrow$ keep phase margin.
Or pick desired L (e.g. $\frac{\alpha}{s}$, α sets ω_{gc}) then $C=L/P$ — avoid cancelling unstable poles/zeros.

Robustness: relative model error $\Delta m(s) = \frac{\Delta G(s)}{G(s)}$; robust stability $\Leftarrow |T(j\omega)| < \frac{1}{|\Delta m(j\omega)|}$ — keep T small where Δm large (high ω).

17. Fundamental Limitations (Lec 11,12)

- (1) $S+T=1$ — seesaw trade-off.
- (2) Gain-phase relation — steep roll-off $\Rightarrow$ phase lag, low PM.
- (3) Interpolation constraints — RHP poles/zeros can't be hidden.
- (4) Bode sensitivity integral (waterbed):

$$\int_0^\infty \log |S(j\omega)| d\omega = \pi \sum_k p_k$$

p_k =unstable open-loop poles; push $|S|$ down somewhere $\Rightarrow$ up elsewhere.

19. Block Diagram Algebra

Series: $G_1 G_2$. **Parallel:** $G_1 \pm G_2$.
Neg. feedback: $\frac{G}{1+GH}$; unity: $\frac{G}{1+G}$.
Pos. feedback: $\frac{G}{1-GH}$.

$L(s) = C(s)P(s)$ (loop TF), $T(s) = \frac{L(s)}{1+L(s)}$, $S(s) = \frac{1}{1+L(s)}$ (sensitivity function maps disturbance on plant output to final output).
Cascaded SS — recipe: EOM of each subsystem:

$$\dot{x}_1 = A_1 x_1 + B_1 u_1, \quad y_1 = C_1 x_1; \quad \dot{x}_2 = A_2 x_2 + B_2 u_2, \quad y_2 = C_2 x_2$$

Sub interconnection $u_2=y_1=C_1 x_1$ into $\dot{x}_2$; stack $x=[x_1; x_2]$:

$$\dot{x} = \begin{bmatrix} A_1 & 0 \\ B_2 C_1 & A_2 \end{bmatrix} x + \begin{bmatrix} B_1 \\ 0 \end{bmatrix} u_1, \quad y_2 = [0 \quad C_2] x$$

TF: get $G_i = C_i(sI - A_i)^{-1}B_i + D_i$ for each subsystem, then $G_{tot} = G_2 G_1$ (multiply) — far easier than inverting the big $(sI - A)$.

20. Quick Reference / Exam Tips

- Stable $\Leftrightarrow$ all poles/eigenvalues in open LHP.
- Reachable: $\det[B \ AB] \neq 0$; Observable: $\det[C \ CA] \neq 0$ (2nd order).
- PD design: pick ζ from M_p , ω_n from t_s , match denom.
- Nyquist through $(\alpha, 0)$: set $G(s) = \alpha$; root is purely imaginary $s = \pm j\omega$.
- Zero SS error tracking: use **integral** action (robust) or feedforward N (needs exact model).
- Real-world uncertainty: parameter errors, disturbances, sensor noise, unmodelled nonlinearity.

PID vs state-feedback: **PID:** model-free, mainly SISO, heuristic/robust, simple to tune, limited pole control. **State-FB:** needs accurate model + full state (or observer), handles MIMO, gives exact pole placement.

Useful: $\zeta=0.5 \Rightarrow M_p \approx 16\%$; $\zeta=0.707 \Rightarrow M_p \approx 4.3\%$; $\zeta=0.6 \Rightarrow M_p \approx 9.5\%$.
 $\sqrt{1-\zeta^2}$ at $\zeta=0.707$ is 0.707.

21. Math Toolbox

2×2 **inverse:** $\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$.
 3×3 **det:** $\det \begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} = a(ei-fh) - b(di-fg) + c(dh-eg)$.
 2×2 **eigenvalues:** $\det(\lambda I - A) = \lambda^2 - \text{tr}(A)\lambda + \det(A)$
Common integrals: $\int e^{-at} dt = -\frac{1}{a} e^{-at}$; $\int_0^\infty e^{-at} dt = \frac{1}{a}$ ($a>0$).
Sin/cos deriv: $\frac{d}{dt} \sin \omega t = \omega \cos \omega t$, $\frac{d}{dt} \cos \omega t = -\omega \sin \omega t$.
Sin/cos integ: $\int \sin \omega t dt = -\frac{1}{\omega} \cos \omega t$, $\int \cos \omega t dt = \frac{1}{\omega} \sin \omega t$.